Notes: Sum and Difference of Cubes, and Factoring by Grouping

Honors Precalculus · Unit 3 (Polynomials and Factoring) · Completed notes for anyone who missed class

Part 1: Sum and Difference of Cubes

Perfect cubes to know: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000. Variable cubes: x^3, $x^6 = (x^2)^3$, $x^9 = (x^3)^3$.

Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Sign memory trick **SOAP**: the binomial keeps the **S**ame sign, the trinomial's middle term has the **O**pposite sign, and the last term is **A**lways **P**ositive.

Why it works: multiply $(a - b)(a^2 + ab + b^2) = a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3 = a^3 - b^3$. The middle terms cancel.

Steps

1. **Take out a GCF first** if there is one.
2. **Find a and b:** take the cube root of each term (for example, $27x^3$ is $(3x)^3$, so $a = 3x$).
3. **Write the binomial** $(a \pm b)$, then the **trinomial** using SOAP: a^2 and b^2 are squared, and the middle term is ab.
4. **Stop.** The trinomial does not factor any further over the real numbers. Check by multiplying if you have time.

Examples

1. $x^3 + 8$ $(a = x, b = 2)$
$= (x + 2)(x^2 - 2x + 4)$

2. $27y^3 - 125$ $(a = 3y, b = 5)$
$= (3y - 5)(9y^2 + 15y + 25)$ (middle term: $3y \cdot 5 = 15y$)

3. $8m^3 + 343$ $(a = 2m, b = 7)$
$= (2m + 7)(4m^2 - 14m + 49)$

4. $54x^3 - 16$ (GCF first!)
$= 2(27x^3 - 8) = 2(3x - 2)(9x^2 + 6x + 4)$

5. $5x^4 + 40x$ (GCF is 5x)
$= 5x(x^3 + 8) = 5x(x + 2)(x^2 - 2x + 4)$

Watch out: $x^3 + 10$ is not a sum of cubes, because 10 is not a perfect cube, so it does not factor this way. And a plain sum of *squares* such as $x^2 + 9$ does not factor over the real numbers.

Part 2: Factoring by Grouping (four terms)

Use grouping when a polynomial has **four terms** and no GCF for all of them. The goal is a **common binomial** that shows up in both groups.

Steps

1. **Group** the terms in pairs: (first two) + (last two).
2. **Factor the GCF out of each pair.** If the third term has a minus sign, factor out a **negative** so the binomials match.
3. The two binomials should be **identical** (the "twins"). Factor the twin out front: (twin)(what is left).
4. **Keep going:** check whether either factor can be factored more (for example, a difference of squares).
5. If the binomials do not match, **rearrange** the terms and try again.

Examples

> **1.** $x^3 + 3x^2 + 5x + 15$
> $= (x^3 + 3x^2) + (5x + 15) = x^2(x + 3) + 5(x + 3) = (x + 3)(x^2 + 5)$

> **2.** $2x^3 - 6x^2 + 5x - 15$
> $= 2x^2(x - 3) + 5(x - 3) = (x - 3)(2x^2 + 5)$

> **3.** $x^3 - 4x^2 - 9x + 36$ (negative GCF, then keep factoring)
> $= x^2(x - 4) - 9(x - 4) = (x - 4)(x^2 - 9) = (x - 4)(x - 3)(x + 3)$

> **4.** $3ax + 3ay + 2bx + 2by$ (variables in every term)
> $= 3a(x + y) + 2b(x + y) = (x + y)(3a + 2b)$

> **5.** $xy + 6 + 2x + 3y$ (rearrange first)
> $= xy + 2x + 3y + 6 = x(y + 2) + 3(y + 2) = (y + 2)(x + 3)$

Check your work: multiply your answer back out. You should get the original four terms.

Check Your Understanding

Try these on your own, then check the answers below. Factor completely.

1. $x^3 - 27$

2. $64a^3 + 1$

3. $250x^3 + 2$

4. $y^3 + 6y^2 + 4y + 24$

5. $3x^3 - 12x^2 - 4x + 16$

6. $x^4 - x$

7. $2x^3 - 8x^2 - 18x + 72$

8. Can $27x^3 + 9$ be factored as a sum of cubes? Explain.

Answers

#	Answer
1	$(x - 3)(x^2 + 3x + 9)$
2	$(4a + 1)(16a^2 - 4a + 1)$
3	$2(125x^3 + 1) = 2(5x + 1)(25x^2 - 5x + 1)$
4	$y^2(y + 6) + 4(y + 6) = (y + 6)(y^2 + 4)$
5	$3x^2(x - 4) - 4(x - 4) = (x - 4)(3x^2 - 4)$
6	$x(x^3 - 1) = x(x - 1)(x^2 + x + 1)$
7	$2x^2(x - 4) - 18(x - 4) = (x - 4)(2x^2 - 18) = 2(x - 4)(x - 3)(x + 3)$
8	No. 9 is not a perfect cube. Take out the GCF 9 to get $9(3x^3 + 1)$, and $3x^3$ is not a perfect cube either, so it is finished.